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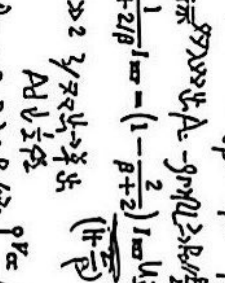
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1. 电流源

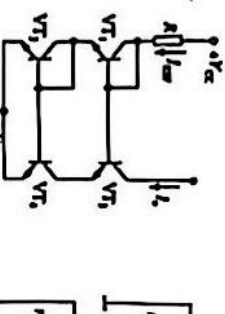
(1) 基本镜像电流源, 如图 7.1 所示

$$I_{C1} = I_{C2} = I_{C3} = \frac{1}{1+2\beta} I_{C4} = (1 - \frac{2}{\beta+2}) I_{C4}$$

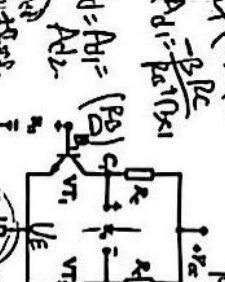
$$I_{C1} \approx I_{C2} = \frac{V_{CC} - U_{BE}}{R} \cdot \beta \approx 2 \cdot \frac{V_{CC} - U_{BE}}{R}$$



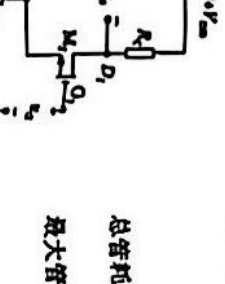
输出电阻: $R_o = r_{ce}$



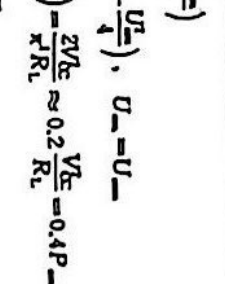
输出电阻: $R_o = r_{ce1} + (1+\beta)(r_{ce2} + R_{E2})$



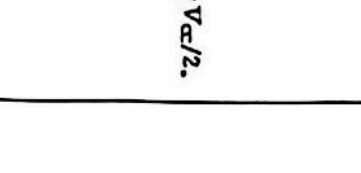
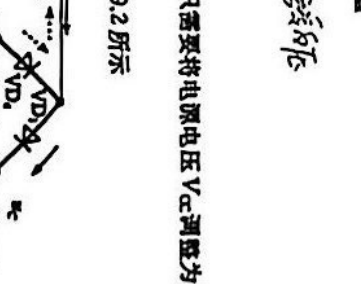
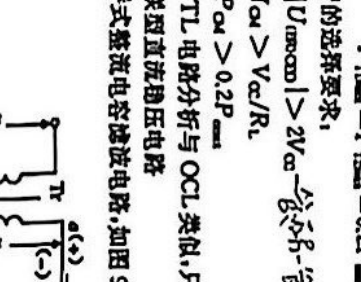
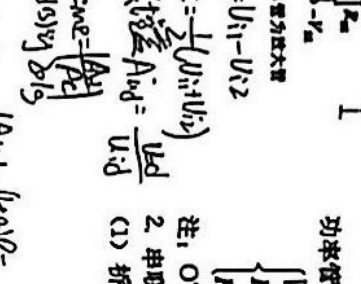
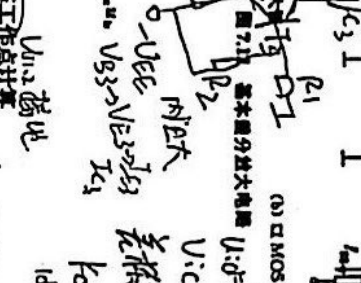
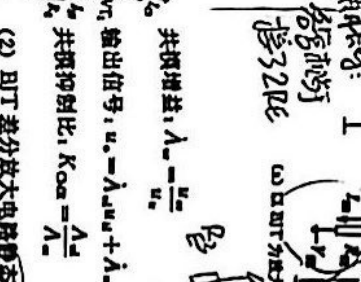
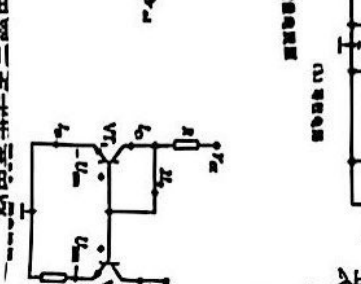
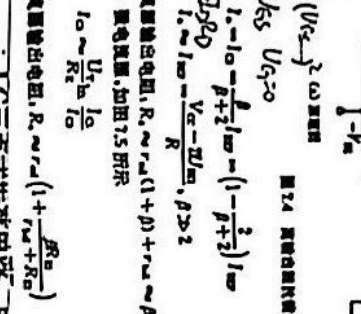
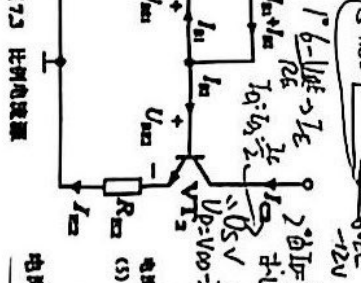
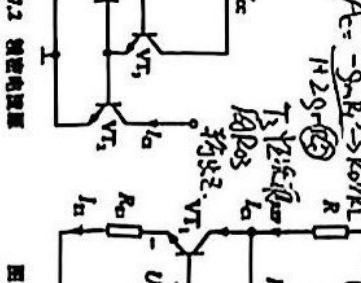
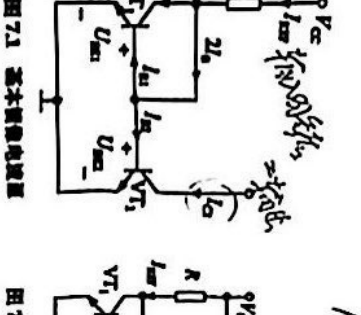
输出电阻: $R_o = r_{ce1} + (1+\beta)(r_{ce2} + R_{E2})$



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(2) 精密电流源, 如图 7.2 所示

$$I_{C1} = I_{C2} = \frac{1}{1+2(\beta+1)} I_{C4} = (1 - \frac{2}{\beta+3}) I_{C4}$$

$$I_{C1} \approx I_{C2} = \frac{V_{CC} - U_{BE}}{R} \cdot \beta \approx 2 \cdot \frac{V_{CC} - U_{BE}}{R}$$

输出电阻: $R_o = r_{ce1} + (1+\beta)(r_{ce2} + R_{E2})$

比例电流源, 如图 7.3 所示

$$I_{C1} = \frac{R_2}{R_1} I_{C2}$$

其中: $I_{C1} \approx \frac{V_{CC} - U_{BE}}{R_1 + R_2} \cdot \frac{R_2}{R_1}$

电流源输出电阻: $R_o = r_{ce1} + (1+\beta)(r_{ce2} + R_{E2})$

(3) 比例电流源, 如图 7.3 所示

$$I_{C1} = \frac{R_2}{R_1} I_{C2}$$

其中: $I_{C1} \approx \frac{V_{CC} - U_{BE}}{R_1 + R_2} \cdot \frac{R_2}{R_1}$

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电流源输出电阻: $R_o = r_{ce1} + (1+\beta)(r_{ce2} + R_{E2})$

比例电流源, 如图 7.3 所示

输入电阻: $R_i = \frac{1}{2} [r_{be} + (1+\beta) \times 2R_E]$

输出电阻: $R_o = 2R_E$

共模放大倍数: $A_{cm} = 0$

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